writeup

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Abstract

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1 Introduction

2 Model Training

In order to examine the representation of binomials in language models, we trained six autoregressive language models from scratch using the OPT architecture (Zhang et al., 2022), crossing two training corpora with three model sizes (125M, 350M, and 1.3B parameters), for a total of six models. All models share a common tokenizer: a Byte-Pair Encoding (BPE) tokenizer with a vocabulary of 8,192 tokens and a context window of 1,024 tokens, trained once on the full BabyLM corpus and reused across all six models to ensure that differences between models reflect training data and scale rather than tokenization. Models were trained on two NVIDIA H100 GPUs using PyTorch Distributed Data Parallel (DDP) in bfloat16 precision, with AdamW optimization and a linear learning rate schedule with 4,000 warmup steps.

The first corpus that we trained the models on was BabyLM(Version 3, Charpentier et al.), a developmentally plausible corpus of approximately 150M tokens consisting of child-directed speech (CHILDES), children's books, and simple web text. Each model was trained for 64 epochs over this corpus, yielding approximately 9.1B total training tokens. The second corpus was a 10B-token subset of C4 (Raffel et al., 2020), a filtered English web text corpus. C4 models were trained for a single pass over this corpus, also yielding approximately 10B training tokens. Comparing the two corpora allows us to examine how the content and diversity of the training data shapes the learning of binomials in language models.

Checkpoints were saved approximately every 10M tokens throughout training and pushed to HuggingFace Hub, giving us dense coverage of each model's learning trajectory. All models and checkpoints will be made publicly available.[1] Table 1 summarizes the key training parameters for each model.

Corpus	Params	Total tokens	Checkpoints	LR
BabyLM	125M	~9.1B	~910	3×10^{-4}
BabyLM	350M	~9.1B	~910	1×10^{-4}
BabyLM	1.3B	~9.1B	~860	1×10^{-4}
C4	125M	~10B	~1,010	3×10^{-4}
C4	350M	~10B	~1,005	1×10^{-4}
C4	1.3B	~10B	~960	1×10^{-4}

Table 1: Training configuration for all six models. All models use the OPT architecture, a shared BPE tokenizer (vocab 8,192; context 1,024 tokens), AdamW optimization, and a linear LR schedule with 4,000 warmup steps.

[1] The huggingface account would reveal the identity of one of the authors and is thus omitted from the double-blind review phase.

All six models showed smooth convergence in training loss throughout training, starting near the theoretical random baseline ($\ln 8192 \approx 9.01$) and decreasing consistently. Held-out validation perplexity confirmed successful training for all models; full loss curves and perplexity figures are reported in Appendix A. Notably, the BabyLM 1.3B model showed elevated final validation perplexity relative to the 350M model (2,557 vs. 290), consistent with mild overfitting given the small corpus (approximately 150M tokens) — unsurprising for a 1.3B parameter model trained on such limited data. We return to the implications of this for our main analyses in the Results.

3 Binomial Ordering Preferences

In this section we present three complementary analyses examining how our models represent binomial ordering preferences over the course of training. The first analysis examines raw **preference** (how strongly each model favors the alphabetically ordered form of a binomial). The second examines the **signed gap** (the extent to which models' preferences deviate from what corpus frequency alone would predict). The third examines **deviation from corpus predictions** as a function of training, examining how the ordering preferences of models deviates from its training distribution as a function of learning.

3.1 Preference

In this section, we examine whether our models learn binomial ordering preferences, and if so what factors predict that preference. For each model checkpoint and binomial pair, we compute the model's ordering **preference** as the difference in log-probability of the alphabetically ordered form (e.g., *bread and butter*) compared to the non-alphabetically ordered form (e.g., *butter and bread*). The alphabetical form is used as a relatively neutral baseline (following Morgan & Levy, 2016a). Concretely, for a binomial with alphabetically ordered token sequence s_α and non-alphabetically ordereed s_n:

$$\text{preference}_i = \log P_\theta(s_{\alpha,i}) - \log P_\theta(s_{n,i})$$

where $\log P_\theta(s)$ is the summed log-probability of sequence s under model parameters θ. Positive values indicate that the model assigns higher probability to the alphabetically ordered form.

We regress preference on three item-level predictors using a Bayesian mixed-effects regression model (Bürkner, 2017). **Generative Preference** (GenPref) is an estimate of ordering preferences driven by a combination of phonological and semantic predictors (Morgan & Levy, 2016a). **RelFreq** is the relative corpus frequency of the alphabetically ordered form (centered):

$$\text{RelFreq}_i = P(\text{alpha})_i - 0.5, \quad \text{where} \quad P(\text{alpha})_i = \frac{c_{\alpha,i}}{c_{\alpha,i} + c_{n,i}}$$

and $c_{\alpha,i}$, $c_{n,i}$ are the corpus counts of the alpha-ordered and non-alphabetical forms of binomial i, respectively. **Log total frequency** is $\ln(c_{\alpha,i} + c_{n,i})$, the natural log of total corpus occurrences of the binomial in either order, centered. We include interaction terms for GenPref and log total frequency as well as RelFreq and log total frequency, allowing the model to capture how ordering preferences may be differently affected by GenPref and RelFreq depending on the overall frequency of that binomial:

$$\text{preference} \sim \text{GenPref} \times \ln(\text{total}) + \text{RelFreq} \times \ln(\text{total}) + (1 \mid \text{binomial})$$

We fit this model at the final training checkpoint and at 20 evenly-spaced checkpoints throughout training to examine the learning trajectory of binomial ordering preferences. We additionally fit a checkpoint-interacted model in which training stage (tokens encountered,

scaled and centered) is included as a predictor to test whether effects grow or shrink linearly over training:

$$\text{preference} \sim (\text{GenPref} + \text{RelFreq} + \ln(\text{total})) \times \text{tokens} + (1 \mid \text{binomial})$$

3.1.1 Results

Final Checkpoint The results of the Bayesian mixed-effects regression at the final checkpoint are presented in Table 3 and visualized in Figure 1. The results suggest that relative frequency is a credible predictor of the model's ordering preferences across all six models, which is consistent with models' ordering preferences being at least partly driven by the distributional statistics of their training data. Interestingly, the effect of GenPref shows a graded pattern across model sizes, with the 125M parameter models showing a credible positive effect, the 350M models showing a somewhat weaker effect, and the 1.3B models showing little credible evidence of a GenPref effect.

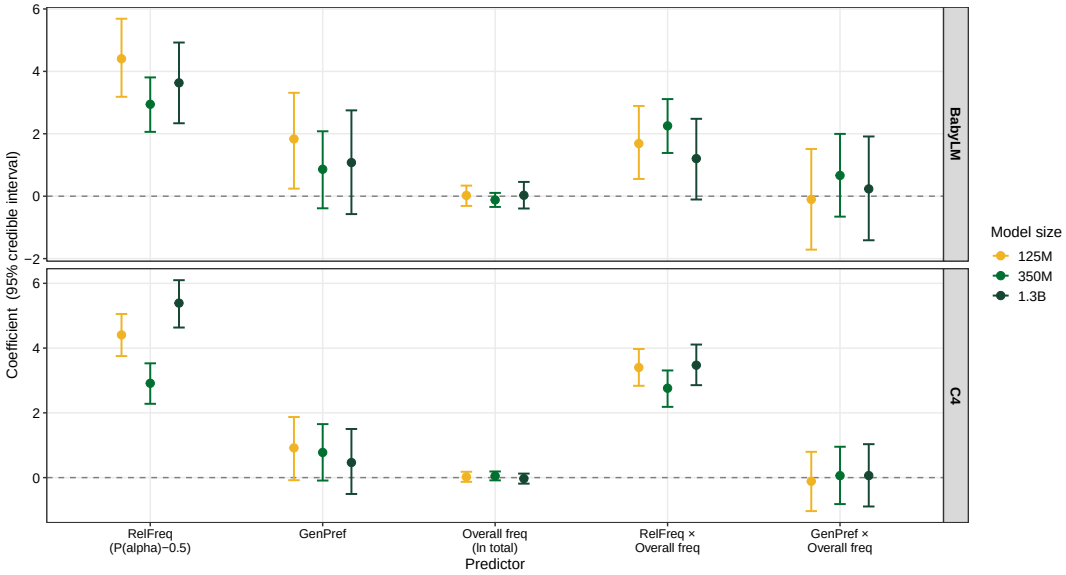

Figure 1: Posterior fixed-effect estimates for predictors of raw preference at the final training checkpoint. Error bars show 95% credible intervals.

GenPref Significance Over Training The results from Figure 2 suggest a size-graded pattern across the BabyLMs, with smaller models showing credible GenPref effects from early in training while larger models show weaker effects throughout training. Interestingly, the C4-trained 1.3B model shows a credible GenPref effect early in training that attenuates with further exposure, suggesting that the learning of generative preferences may depend on an interaction between model scale and the volume and diversity of the training data.

Checkpoint-Interacted Model The results of the checkpoint-interacted model are presented in Table 4 and visualized in Figure 3. Because all predictors are centered, main effects in this model reflect the estimated influence of each predictor at the midpoint of training. In other words, the main effects of GenPref and RelFreq reflect their respective influences halfway through training, while interactions with training stage indicate whether each effect grows or shrinks linearly over the course of training.

The results suggest that there is an effect of relative frequency across all six models at the training midpoint, and that this effect grows stronger as training progresses. Interestingly, the GenPref results differ across corpora and model sizes. Among the BabyLM models, the results suggest that only the 125M model shows an effect of GenPref at the training midpoint, and that this effect grows further over the course of training, while the 350M

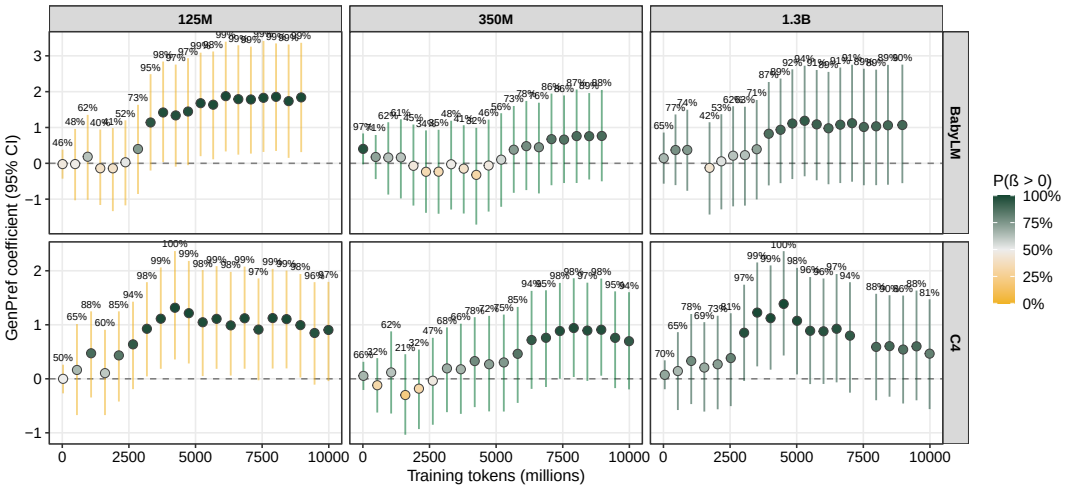

Figure 2: GenPref coefficient (95% CI) at each of 20 sampled training checkpoints, for raw preference models. Point fill encodes $P(\beta > 0)$: green indicates the posterior is predominantly positive, gold predominantly negative, grey uncertain.

and 1.3B BabyLM models show little credible evidence of a GenPref effect. Among the C4 models, the results suggest that all three models show a credible GenPref effect at the training midpoint, though this effect appears to be weakest for the 350M model.

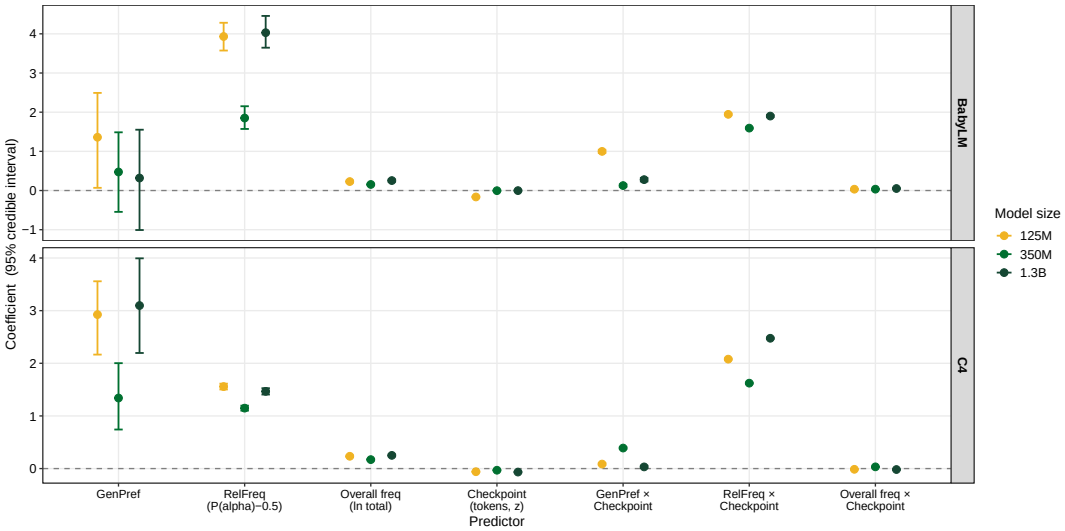

Figure 3: Posterior fixed-effect estimates from the checkpoint-interacted preference model. Interactions with training stage (tokens, z-scored) test whether each predictor's effect on raw preference changes linearly over the course of training.

3.2 Signed Gap

Since we have access to the exact training data each model encountered, we are able to calculate exactly how many times the model has seen each ordering of each binomial at each checkpoint. Thus, in this section, we examine the **signed gap**: the difference between each model's preference score and the log corpus frequency ratio for that binomial. Formally:

$$\text{signed gap}_i = \text{preference}_i - \ln \frac{c_{\alpha,i}}{c_{n,i}}$$

where preference$_i$ is defined as above and $c_{\alpha,i}/c_{n,i}$ is the corpus frequency ratio of the alphabetically ordered to the non-alphabetical form. A signed gap of zero means the model's preference exactly matches what corpus frequency predicts; positive values indicate a stronger preference for alpha order than corpus statistics would suggest, and negative values the reverse. Using the signed gap as the outcome allows us to ask whether models develop ordering preferences that go beyond simply reflecting the distributional statistics of their training data. We regress signed gap on the same predictors as the preference model:

$$\text{signed gap} \sim \text{GenPref} \times \ln(\text{total}) + \text{RelFreq} \times \ln(\text{total}) + (1 \mid \text{binomial})$$

We fit this model at the final checkpoint and at 20 evenly-spaced checkpoints throughout training. We additionally fit a checkpoint-interacted model to test whether the effect of each predictor grows or shrinks linearly over training:

$$\text{signed gap} \sim (\text{GenPref} + \text{RelFreq} + \ln(\text{total})) \times \text{tokens} + (1 \mid \text{binomial})$$

3.2.1 Results

Final Checkpoint The results of the Bayesian mixed-effects regression for signed gap at the final checkpoint are presented in Table 5 and visualized in Figure 4. Recall that because the signed gap removes the contribution of corpus frequency from the preference score, a credible positive effect of GenPref here would suggest that models develop ordering preferences consistent with phonological and semantic biases over and above what the training corpus frequency ratio would predict. A positive effect of relative frequency, on the other hand, would indicate that the model's ordering preference for the alphabetically ordered form is more extreme than the training data alone would predict, while a negative effect would indicate the opposite.

The results for relative frequency are quite mixed across model sizes and corpora. Among the BabyLM models, the largest model shows a negative effect of relative frequency, while the two smaller models show little to no effect. Among the C4 models, the pattern is reversed: the two smaller models show a negative effect of relative frequency, while the largest shows little evidence of an effect. In terms of generative preferences, the 125M BabyLM model shows a weak positive effect of GenPref, while the two larger BabyLM models show little evidence of an effect. For the C4 models, the pattern is similar: the 125M C4 model shows a possible effect of GenPref (approximately 89% of posterior samples were greater than zero), while the two larger C4 models show little evidence of an effect.

Taken together, the signed gap results suggest that models' ordering preferences are broadly consistent with what corpus frequency predicts, with limited evidence of preferences that go beyond the training distribution. Interestingly, the modest GenPref effects that do appear are consistently concentrated in the smallest models across both corpora, suggesting that smaller models may be more sensitive to abstract ordering biases even when controlling for frequency.

GenPref Significance Over Training The results from Figure 5 largely mirror the preference trajectory in Figure 2. The 125M models show a steady increase in the GenPref coefficient over training regardless of corpus, while the two larger BabyLM models show little to no effect of GenPref throughout. Interestingly, the two larger C4 models do show some evidence of a GenPref effect over training, suggesting that training data diversity may play a role in the learning of generative preferences. Additionally, as was the case in the preference trajectory, the largest C4 model shows a U-shaped pattern, with the GenPref effect initially growing before subsequently declining.

Checkpoint-Interacted Model The results of the checkpoint-interacted signed gap model are presented in Table 6 and visualized in Figure 6. Among the BabyLM models, the results suggest that the 125M model shows a credible effect of GenPref at the training midpoint, and that this effect grows stronger as training progresses. The negative effect of relative frequency across all three BabyLM models suggests that, at the midpoint of training, the

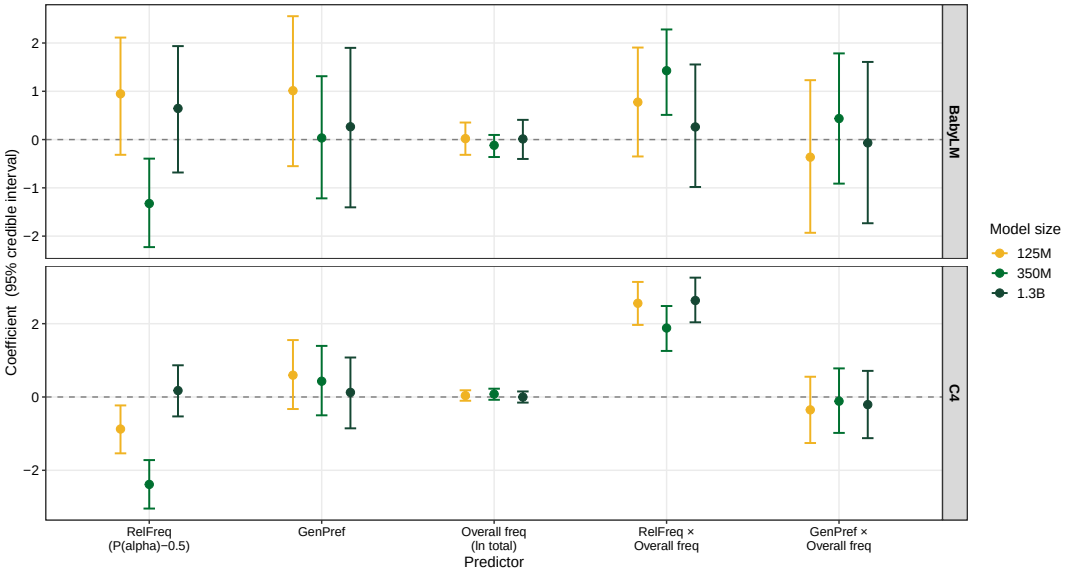

Figure 4: Posterior fixed-effect estimates for predictors of signed gap (preference minus log corpus frequency ratio) at the final training checkpoint. Error bars show 95% credible intervals.

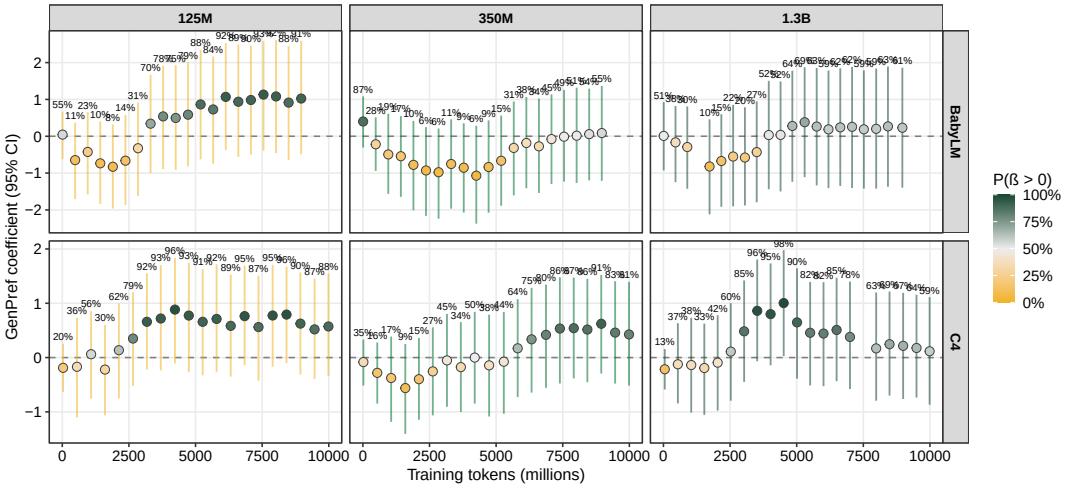

Figure 5: GenPref coefficient (95% CI) at each of 20 sampled training checkpoints, for signed gap models. Point fill encodes $P(\beta > 0)$: green indicates the posterior is predominantly positive, gold predominantly negative, grey uncertain.

models' preferences for the alphabetically ordered form fall short of what corpus frequency alone would predict, and that this gap narrows over the course of training (given the positive interaction effect between relative frequency and checkpoint). For the C4 models, the 125M and 1.3B models show an effect of GenPref, while the 350M model does not. The C4 models show a similar pattern for relative frequency as the BabyLM models, with a negative effect that also narrows over training.

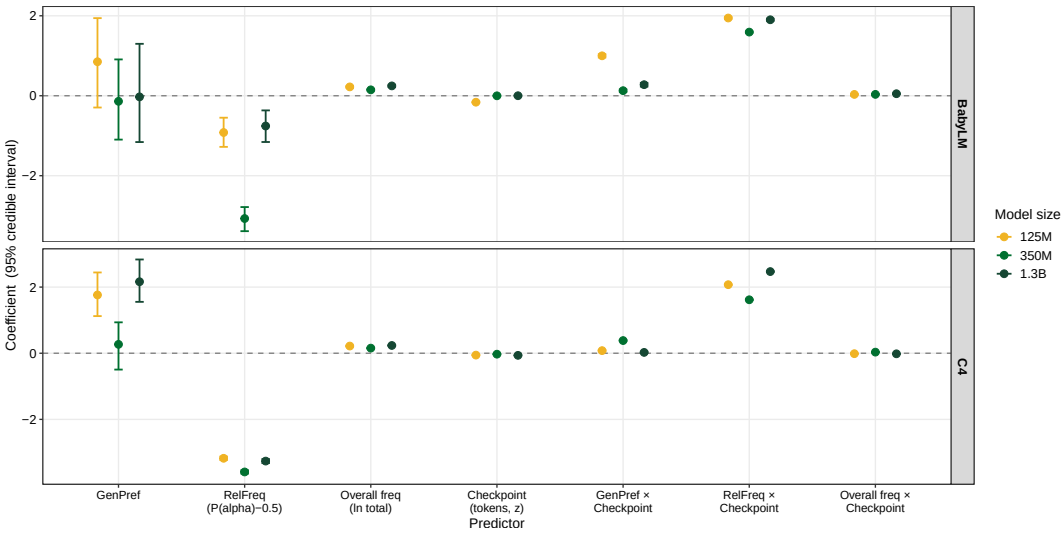

Figure 6: Posterior fixed-effect estimates from the checkpoint-interacted signed gap model.

3.3 Deviation from Corpus Predictions

In this section, we examine how closely models track corpus-frequency predictions over the course of training, using two descriptive summaries and one regression model. First, we compute the **mean absolute signed gap** across all binomials at each checkpoint (Figure 7):

$$\overline{|\text{signed gap}|}_t = \frac{1}{N}\sum_{i=1}^{N}\left|\text{signed gap}_{i,t}\right|$$

where t indexes the checkpoint and N is the number of binomials. Lower values indicate that models' preferences are, on average, closer to what corpus frequency would predict. Second, we compute the **Pearson correlation between model preference and log corpus frequency ratio** across items at each checkpoint (Figure 8):

$$r_t = \text{Cor}\left(\text{preference}_{i,t},\ \ln\frac{c_{\alpha,i}}{c_{n,i}}\right)$$

Values near 1 indicate near-perfect frequency tracking. Finally, we fit a model predicting the absolute signed gap from training stage and log total frequency to test whether models become more or less corpus-like as training progresses and whether this depends on item frequency (Figure 9; Table 7):

$$|\text{signed gap}| \sim \text{tokens} \times \ln(\text{total}) + (1 \mid \text{binomial})$$

3.3.1 Results

The results from Figure 7 suggest that the mean absolute signed gap grows throughout training for all six models, indicating that models' ordering preferences deviate more from

what corpus frequency alone would predict as training progresses. Among the BabyLM models, this growth is most pronounced for the 1.3B model and weakest for the 350M model. Among the C4 models, all three follow a broadly similar trajectory.

Interestingly, Figure 8 suggests that even as the absolute signed gap grows over training, models' ordering preferences simultaneously become more correlated with corpus frequency. In other words, models' preferences grow more extreme over training, but in a direction that is broadly consistent with the corpus frequency ordering.

Finally, the results of the Bayesian mixed-effects regression are presented in Table 7 and visualized in Figure 9. The effect of training stage on absolute signed gap is quite mixed across models. Among the BabyLM models, the 125M and 1.3B models show a positive effect, while the 350M model shows a negative effect. Among the C4 models, the two smaller models show a slightly negative effect while the largest shows a slightly positive effect. In general, the effect of training stage appears weaker for the C4 models than for the BabyLM models, suggesting that training data diversity may dampen the growth of the gap between models' learned preferences and what corpus frequency alone predicts.

The effect of overall frequency also varies across models. The 125M and 1.3B BabyLM models show a slightly negative effect of overall frequency, suggesting that the absolute signed gap is smaller for more frequently encountered binomials, while the 350M BabyLM model shows a large positive effect, suggesting the opposite. All three C4 models show a strong positive effect of overall frequency, indicating that the gap between model preference and the frequency-based prediction is larger for higher-frequency binomials. This pattern is of particular interest given existing work demonstrating frequency-dependent preference extremity in humans, whereby humans develop more extreme ordering preferences for high-frequency binomials over time (e.g., Morgan & Levy, 2016b; Houghton & Morgan, 2024; Houghton, 2025). The present results suggest that model size and training data diversity may modulate a similar effect in language models, though a full investigation of this question is left for future research.

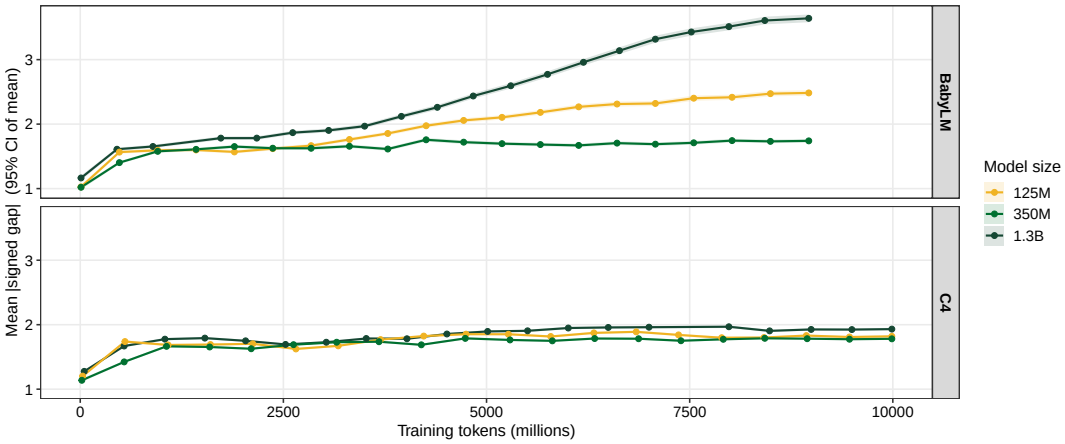

Figure 7: Mean absolute signed gap across items at each of 20 sampled checkpoints. Higher values indicate that models' preference magnitudes deviate more from what corpus frequency predicts, on average.

4 Discussion

The results of the present study suggest that the models trained here do develop binomial ordering preferences over the course of training, and that these preferences are at least partly driven by the distributional statistics of the training data. Beyond frequency, however, the picture is more nuanced. Smaller models — particularly the 125M models across both corpora — show the most consistent evidence of GenPref effects, suggesting that they are sensitive to abstract ordering preferences over and above what corpus frequency alone

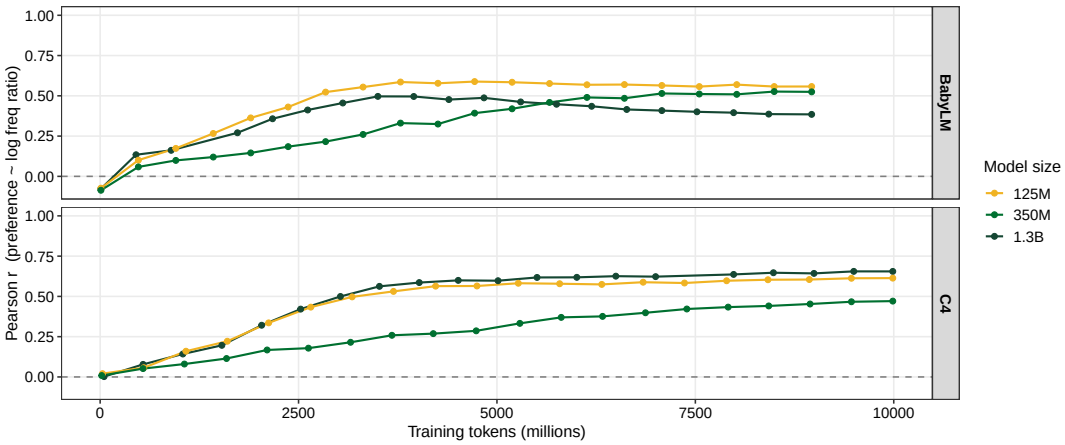

Figure 8: Pearson correlation between model preference and log corpus frequency ratio across items at each sampled checkpoint. Values near 1 indicate that models rank binomials by preference in the same order as the corpus frequency ratio.

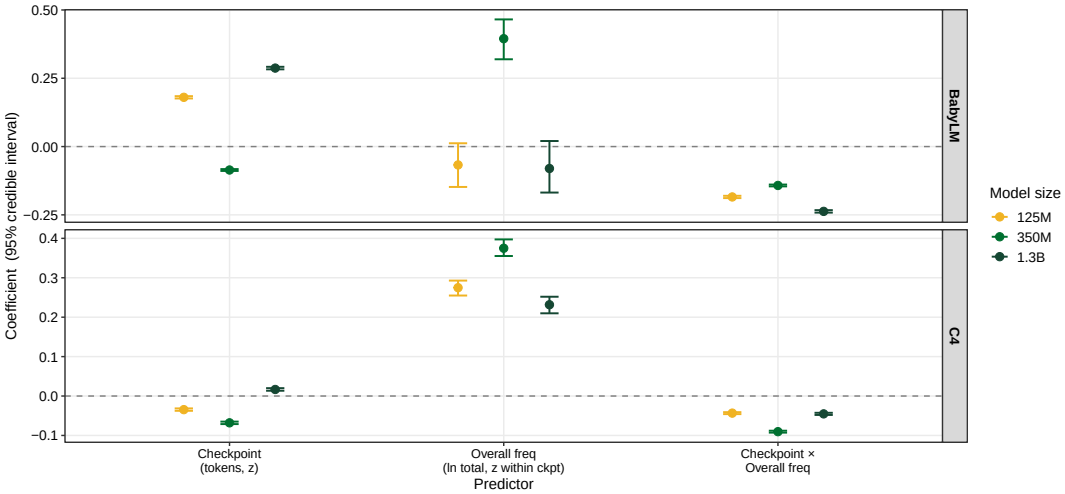

Figure 9: Posterior fixed-effect estimates for predictors of absolute signed gap. The main effect of training stage (tokens, z) captures overall growth or shrinkage in preference magnitude; the interaction tests whether this depends on item-level corpus exposure.

would predict. Larger models, by contrast, show weaker or absent GenPref effects at the final checkpoint, though the trajectory results suggest that these models do show some sensitivity to GenPref earlier in training, with the effect weakening as training progresses. This may reflect a tendency for larger models to gradually shift toward relying on distributional frequency as they accumulate more training experience, at the expense of abstract knowledge. The nature of the training corpus also appears to matter: C4-trained models generally show weaker deviations from corpus frequency predictions than BabyLM-trained models, suggesting that training data diversity plays a role in shaping models' ordering preferences. Finally, as training progresses, models' preferences grow more extreme in a direction broadly consistent with corpus frequency, but the absolute gap between model preferences and frequency-based predictions also grows, with the size of this gap modulated by both model size and item frequency in a manner reminiscent of frequency-dependent preference extremity observed in humans.

5 Conclusion

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A Training Quality

Corpus	Params	Initial PPL	Final PPL
BabyLM	125M	9725	537.50
BabyLM	350M	9470	289.68
BabyLM	1.3B	13078	2557.08
C4	125M	9528	31.63
C4	350M	9352	43.16
C4	1.3B	12275	27.87

Table 2: Perplexity at initialization and final checkpoint for all six models. Perplexity = e^{loss}; initial perplexity is derived from the first logged training step; final perplexity is measured on a held-out validation set (first 5% of BabyLM training split for BabyLM models; official C4 validation split for C4 models). The BabyLM 1.3B model's elevated final perplexity relative to 350M reflects overfitting pressure on a small corpus.

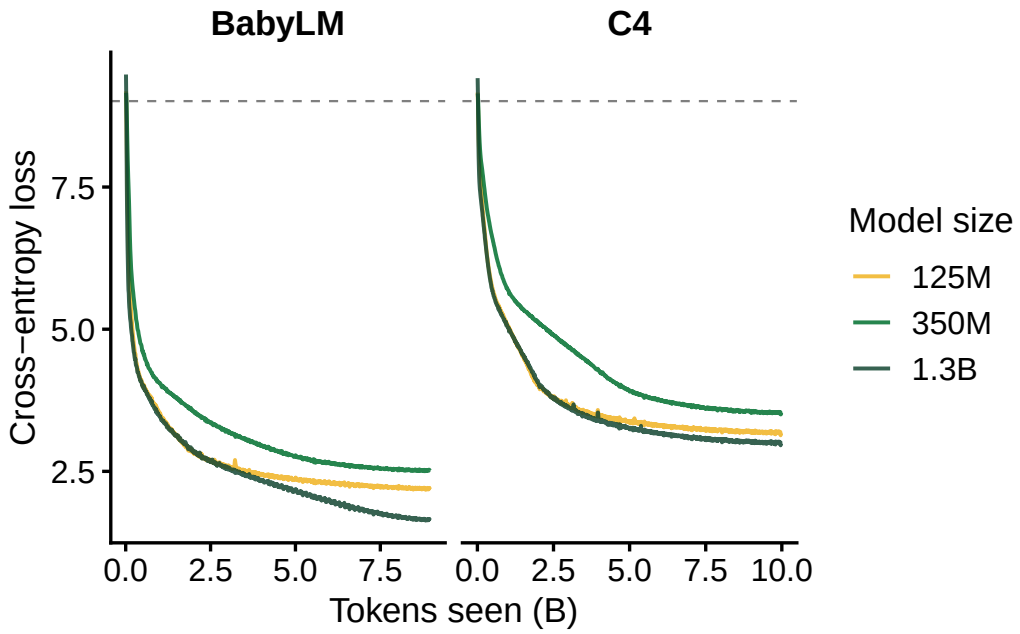

Figure 10: Training loss over the course of training for all six models, separated by corpus. All models start near the theoretical random baseline ($\ln 8192 \approx 9.01$) and converge smoothly; the 1.3B model achieves the lowest final loss in both corpora.

B Posterior Coefficient Tables

B.1 Preference: Final Checkpoint

Corpus	Size	Term	Mean	SD	2.5%	97.5%	$P(\beta > 0)$
BabyLM	125M	GenPref	1.835	0.790	0.243	3.313	0.990
BabyLM	125M	GenPref × Overall freq	-0.105	0.827	-1.713	1.514	0.449
BabyLM	125M	Overall freq (ln total)	0.024	0.170	-0.312	0.339	0.556
BabyLM	125M	RelFreq × Overall freq	1.688	0.587	0.553	2.892	0.998
BabyLM	125M	RelFreq (P(alpha)$-$0.5)	4.404	0.638	3.185	5.687	1.000
BabyLM	350M	GenPref	0.863	0.643	-0.388	2.080	0.910
BabyLM	350M	GenPref × Overall freq	0.664	0.675	-0.655	1.996	0.838
BabyLM	350M	Overall freq (ln total)	-0.120	0.117	-0.344	0.106	0.152
BabyLM	350M	RelFreq × Overall freq	2.254	0.442	1.385	3.112	1.000

Corpus	Size	Term	Mean	SD	2.5%	97.5%	$P(\beta > 0)$
BabyLM	350M	RelFreq (P(alpha)$-$0.5)	2.945	0.447	2.062	3.806	1.000
BabyLM	1.3B	GenPref	1.078	0.848	-0.573	2.750	0.898
BabyLM	1.3B	GenPref × Overall freq	0.234	0.854	-1.412	1.914	0.608
BabyLM	1.3B	Overall freq (ln total)	0.029	0.217	-0.394	0.458	0.554
BabyLM	1.3B	RelFreq × Overall freq	1.205	0.659	-0.106	2.480	0.966
BabyLM	1.3B	RelFreq (P(alpha)$-$0.5)	3.632	0.666	2.337	4.924	1.000
C4	125M	GenPref	0.917	0.493	-0.080	1.872	0.969
C4	125M	GenPref × Overall freq	-0.113	0.465	-1.034	0.796	0.404
C4	125M	Overall freq (ln total)	0.021	0.078	-0.131	0.180	0.607
C4	125M	RelFreq × Overall freq	3.401	0.294	2.835	3.972	1.000
C4	125M	RelFreq (P(alpha)$-$0.5)	4.406	0.336	3.754	5.054	1.000
C4	350M	GenPref	0.777	0.452	-0.091	1.652	0.957
C4	350M	GenPref × Overall freq	0.061	0.452	-0.817	0.953	0.553
C4	350M	Overall freq (ln total)	0.050	0.071	-0.085	0.190	0.761
C4	350M	RelFreq × Overall freq	2.760	0.287	2.185	3.309	1.000
C4	350M	RelFreq (P(alpha)$-$0.5)	2.913	0.319	2.279	3.530	1.000
C4	1.3B	GenPref	0.466	0.509	-0.505	1.502	0.820
C4	1.3B	GenPref × Overall freq	0.063	0.487	-0.892	1.033	0.552
C4	1.3B	Overall freq (ln total)	-0.031	0.080	-0.186	0.124	0.351
C4	1.3B	RelFreq × Overall freq	3.472	0.323	2.855	4.109	1.000
C4	1.3B	RelFreq (P(alpha)$-$0.5)	5.389	0.375	4.635	6.097	1.000

Table 3: Posterior fixed effects for preference at the final checkpoint. Mean = posterior mean; SD = posterior SD; 2.5%/97.5% = 95% credible interval bounds.

B.2 Preference: Checkpoint-Interacted Model

Corpus	Size	Term	Mean	SD	2.5%	97.5%	$P(\beta > 0)$
BabyLM	125M	Checkpoint (tokens, z)	-0.164	0.006	-0.177	-0.152	0.000
BabyLM	125M	GenPref	1.360	0.618	0.069	2.492	0.986
BabyLM	125M	GenPref × Checkpoint	0.999	0.021	0.958	1.040	1.000
BabyLM	125M	Overall freq × Checkpoint	0.034	0.003	0.028	0.040	1.000
BabyLM	125M	Overall freq (ln total)	0.229	0.013	0.204	0.254	1.000
BabyLM	125M	RelFreq × Checkpoint	1.945	0.012	1.922	1.968	1.000
BabyLM	125M	RelFreq (P(alpha)$-$0.5)	3.932	0.183	3.574	4.282	1.000
BabyLM	350M	Checkpoint (tokens, z)	-0.004	0.005	-0.013	0.005	0.209
BabyLM	350M	GenPref	0.474	0.522	-0.545	1.486	0.818
BabyLM	350M	GenPref × Checkpoint	0.126	0.016	0.095	0.157	1.000
BabyLM	350M	Overall freq × Checkpoint	0.035	0.002	0.030	0.039	1.000
BabyLM	350M	Overall freq (ln total)	0.155	0.010	0.136	0.174	1.000
BabyLM	350M	RelFreq × Checkpoint	1.593	0.009	1.576	1.611	1.000
BabyLM	350M	RelFreq (P(alpha)$-$0.5)	1.851	0.146	1.572	2.152	1.000
BabyLM	1.3B	Checkpoint (tokens, z)	-0.002	0.007	-0.016	0.012	0.372
BabyLM	1.3B	GenPref	0.319	0.662	-1.008	1.552	0.685
BabyLM	1.3B	GenPref × Checkpoint	0.278	0.023	0.232	0.323	1.000
BabyLM	1.3B	Overall freq × Checkpoint	0.051	0.003	0.044	0.058	1.000
BabyLM	1.3B	Overall freq (ln total)	0.254	0.014	0.225	0.282	1.000
BabyLM	1.3B	RelFreq × Checkpoint	1.901	0.013	1.873	1.926	1.000
BabyLM	1.3B	RelFreq (P(alpha)$-$0.5)	4.030	0.201	3.646	4.457	1.000

Corpus	Size	Term	Mean	SD	2.5%	97.5%	$P(\beta > 0)$
C4	125M	Checkpoint (tokens, z)	-	0.003	-	-	0.000
			0.060		0.066	0.053	
C4	125M	GenPref	2.925	0.373	2.165	3.558	1.000
C4	125M	GenPref × Checkpoint	0.085	0.010	0.066	0.105	1.000
C4	125M	Overall freq × Checkpoint	-	0.001	-	-	0.000
			0.014		0.017	0.011	
C4	125M	Overall freq (ln total)	0.234	0.008	0.217	0.249	1.000
C4	125M	RelFreq × Checkpoint	2.079	0.006	2.066	2.091	1.000
C4	125M	RelFreq (P(alpha)$-$0.5)	1.557	0.028	1.503	1.613	1.000
C4	350M	Checkpoint (tokens, z)	-	0.003	-	-	0.000
			0.031		0.037	0.026	
C4	350M	GenPref	1.340	0.335	0.742	2.002	1.000
C4	350M	GenPref × Checkpoint	0.390	0.009	0.372	0.407	1.000
C4	350M	Overall freq × Checkpoint	0.033	0.001	0.030	0.035	1.000
C4	350M	Overall freq (ln total)	0.170	0.007	0.156	0.184	1.000
C4	350M	RelFreq × Checkpoint	1.622	0.006	1.611	1.633	1.000
C4	350M	RelFreq (P(alpha)$-$0.5)	1.148	0.024	1.101	1.195	1.000
C4	1.3B	Checkpoint (tokens, z)	-	0.004	-	-	0.000
			0.066		0.073	0.060	
C4	1.3B	GenPref	3.097	0.446	2.196	3.993	1.000
C4	1.3B	GenPref × Checkpoint	0.032	0.011	0.010	0.054	0.997
C4	1.3B	Overall freq × Checkpoint	-	0.002	-	-	0.000
			0.018		0.021	0.015	
C4	1.3B	Overall freq (ln total)	0.251	0.009	0.233	0.268	1.000
C4	1.3B	RelFreq × Checkpoint	2.475	0.007	2.461	2.489	1.000
C4	1.3B	RelFreq (P(alpha)$-$0.5)	1.467	0.031	1.407	1.526	1.000

Table 4: Posterior fixed effects from the checkpoint-interacted preference model.

B.3 Signed Gap: Final Checkpoint

Corpus	Size	Term	Mean	SD	2.5%	97.5%	$P(\beta > 0)$
BabyLM	125M	GenPref	1.013	0.796	-0.551	2.555	0.898
BabyLM	125M	GenPref × Overall freq	-0.364	0.808	-1.931	1.230	0.326
BabyLM	125M	Overall freq (ln total)	0.020	0.171	-0.316	0.353	0.546
BabyLM	125M	RelFreq × Overall freq	0.775	0.579	-0.350	1.906	0.910
BabyLM	125M	RelFreq (P(alpha)$-$0.5)	0.948	0.612	-0.316	2.113	0.939
BabyLM	350M	GenPref	0.033	0.642	-1.218	1.312	0.521
BabyLM	350M	GenPref × Overall freq	0.436	0.683	-0.912	1.785	0.738
BabyLM	350M	Overall freq (ln total)	-0.119	0.116	-0.362	0.096	0.153
BabyLM	350M	RelFreq × Overall freq	1.427	0.444	0.511	2.281	0.999
BabyLM	350M	RelFreq (P(alpha)$-$0.5)	-1.325	0.469	-2.228	-0.395	0.002
BabyLM	1.3B	GenPref	0.265	0.842	-1.404	1.900	0.624
BabyLM	1.3B	GenPref × Overall freq	-0.069	0.848	-1.733	1.608	0.468
BabyLM	1.3B	Overall freq (ln total)	0.013	0.209	-0.402	0.408	0.525
BabyLM	1.3B	RelFreq × Overall freq	0.262	0.651	-0.982	1.555	0.656
BabyLM	1.3B	RelFreq (P(alpha)$-$0.5)	0.645	0.663	-0.682	1.936	0.835
C4	125M	GenPref	0.595	0.475	-0.328	1.553	0.895
C4	125M	GenPref × Overall freq	-0.351	0.462	-1.256	0.552	0.224
C4	125M	Overall freq (ln total)	0.039	0.073	-0.103	0.183	0.700
C4	125M	RelFreq × Overall freq	2.557	0.304	1.966	3.139	1.000
C4	125M	RelFreq (P(alpha)$-$0.5)	-0.874	0.334	-1.537	-0.230	0.004
C4	350M	GenPref	0.430	0.478	-0.500	1.394	0.816
C4	350M	GenPref × Overall freq	-0.115	0.456	-0.980	0.780	0.401
C4	350M	Overall freq (ln total)	0.079	0.078	-0.076	0.226	0.847

Corpus	Size	Term	Mean	SD	2.5%	97.5%	$P(\beta > 0)$
C4	350M	RelFreq × Overall freq	1.881	0.313	1.256	2.482	1.000
C4	350M	RelFreq (P(alpha)$-$0.5)	-2.385	0.340	-3.043	-1.721	0.000
C4	1.3B	GenPref	0.125	0.488	-0.855	1.077	0.601
C4	1.3B	GenPref × Overall freq	-0.207	0.466	-1.123	0.713	0.328
C4	1.3B	Overall freq (ln total)	-0.002	0.077	-0.153	0.151	0.489
C4	1.3B	RelFreq × Overall freq	2.632	0.312	2.038	3.255	1.000
C4	1.3B	RelFreq (P(alpha)$-$0.5)	0.175	0.355	-0.530	0.865	0.689

Table 5: Posterior fixed effects for signed gap at the final checkpoint.

B.4 Signed Gap: Checkpoint-Interacted Model

Corpus	Size	Term	Mean	SD	2.5%	97.5%	$P(\beta > 0)$
BabyLM	125M	Checkpoint (tokens, z)	-0.161	0.006	-0.174	-0.149	0.000
BabyLM	125M	GenPref	0.851	0.596	-0.294	1.941	0.923
BabyLM	125M	GenPref × Checkpoint	0.999	0.021	0.958	1.040	1.000
BabyLM	125M	Overall freq × Checkpoint	0.033	0.003	0.027	0.039	1.000
BabyLM	125M	Overall freq (ln total)	0.222	0.013	0.197	0.247	1.000
BabyLM	125M	RelFreq × Checkpoint	1.945	0.012	1.921	1.968	1.000
BabyLM	125M	RelFreq (P(alpha)$-$0.5)	-0.919	0.190	-1.279	-0.549	0.000
BabyLM	350M	Checkpoint (tokens, z)	-0.001	0.005	-0.011	0.008	0.392
BabyLM	350M	GenPref	-0.140	0.500	-1.097	0.909	0.390
BabyLM	350M	GenPref × Checkpoint	0.127	0.016	0.097	0.158	1.000
BabyLM	350M	Overall freq × Checkpoint	0.034	0.002	0.030	0.039	1.000
BabyLM	350M	Overall freq (ln total)	0.148	0.010	0.129	0.167	1.000
BabyLM	350M	RelFreq × Checkpoint	1.592	0.009	1.575	1.609	1.000
BabyLM	350M	RelFreq (P(alpha)$-$0.5)	-3.071	0.151	-3.387	-2.784	0.000
BabyLM	1.3B	Checkpoint (tokens, z)	0.001	0.007	-0.013	0.015	0.543
BabyLM	1.3B	GenPref	-0.028	0.633	-1.158	1.299	0.483
BabyLM	1.3B	GenPref × Checkpoint	0.277	0.024	0.230	0.323	1.000
BabyLM	1.3B	Overall freq × Checkpoint	0.051	0.003	0.044	0.057	1.000
BabyLM	1.3B	Overall freq (ln total)	0.247	0.014	0.218	0.275	1.000
BabyLM	1.3B	RelFreq × Checkpoint	1.900	0.014	1.874	1.927	1.000
BabyLM	1.3B	RelFreq (P(alpha)$-$0.5)	-0.757	0.202	-1.157	-0.366	0.000
C4	125M	Checkpoint (tokens, z)	-0.057	0.003	-0.063	-0.051	0.000
C4	125M	GenPref	1.762	0.360	1.123	2.442	1.000
C4	125M	GenPref × Checkpoint	0.079	0.010	0.059	0.098	1.000
C4	125M	Overall freq × Checkpoint	-0.014	0.001	-0.017	-0.011	0.000
C4	125M	Overall freq (ln total)	0.217	0.009	0.200	0.234	1.000
C4	125M	RelFreq × Checkpoint	2.073	0.006	2.060	2.085	1.000
C4	125M	RelFreq (P(alpha)$-$0.5)	-3.176	0.027	-3.229	-3.124	0.000
C4	350M	Checkpoint (tokens, z)	-0.029	0.003	-0.034	-0.024	0.000

Corpus	Size	Term	Mean	SD	2.5%	97.5%	$P(\beta > 0)$
C4	350M	GenPref	0.268	0.385	-0.494	0.934	0.757
C4	350M	GenPref × Checkpoint	0.383	0.009	0.365	0.401	1.000
C4	350M	Overall freq × Checkpoint	0.033	0.001	0.031	0.036	1.000
C4	350M	Overall freq (ln total)	0.154	0.007	0.141	0.168	1.000
C4	350M	RelFreq × Checkpoint	1.616	0.006	1.605	1.628	1.000
C4	350M	RelFreq (P(alpha)$-$0.5)	-3.590	0.026	-3.639	-3.539	0.000
C4	1.3B	Checkpoint (tokens, z)	-0.064	0.004	-0.071	-0.057	0.000
C4	1.3B	GenPref	2.162	0.345	1.553	2.834	1.000
C4	1.3B	GenPref × Checkpoint	0.025	0.011	0.003	0.048	0.987
C4	1.3B	Overall freq × Checkpoint	-0.018	0.002	-0.021	-0.014	0.000
C4	1.3B	Overall freq (ln total)	0.235	0.009	0.217	0.253	1.000
C4	1.3B	RelFreq × Checkpoint	2.469	0.007	2.455	2.483	1.000
C4	1.3B	RelFreq (P(alpha)$-$0.5)	-3.261	0.031	-3.321	-3.199	0.000

Table 6: Posterior fixed effects from the checkpoint-interacted signed gap model.

B.5 Absolute Signed Gap

Corpus	Size	Term	Mean	SD	2.5%	97.5%	$P(\beta > 0)$
BabyLM	125M	Checkpoint × Overall freq	-0.184	0.002	-0.189	-0.180	0.000
BabyLM	125M	Checkpoint (tokens, z)	0.180	0.002	0.176	0.184	1.000
BabyLM	125M	Overall freq (ln total, z within ckpt)	-0.067	0.040	-0.148	0.012	0.048
BabyLM	350M	Checkpoint × Overall freq	-0.142	0.002	-0.146	-0.139	0.000
BabyLM	350M	Checkpoint (tokens, z)	-0.086	0.002	-0.089	-0.082	0.000
BabyLM	350M	Overall freq (ln total, z within ckpt)	0.395	0.038	0.320	0.465	1.000
BabyLM	1.3B	Checkpoint × Overall freq	-0.237	0.002	-0.242	-0.233	0.000
BabyLM	1.3B	Checkpoint (tokens, z)	0.287	0.002	0.283	0.292	1.000
BabyLM	1.3B	Overall freq (ln total, z within ckpt)	-0.080	0.047	-0.168	0.020	0.044
C4	125M	Checkpoint × Overall freq	-0.043	0.001	-0.046	-0.041	0.000
C4	125M	Checkpoint (tokens, z)	-0.035	0.002	-0.037	-0.031	0.000
C4	125M	Overall freq (ln total, z within ckpt)	0.275	0.010	0.255	0.293	1.000
C4	350M	Checkpoint × Overall freq	-0.090	0.001	-0.093	-0.088	0.000
C4	350M	Checkpoint (tokens, z)	-0.068	0.002	-0.071	-0.065	0.000
C4	350M	Overall freq (ln total, z within ckpt)	0.375	0.011	0.355	0.397	1.000
C4	1.3B	Checkpoint × Overall freq	-0.045	0.001	-0.048	-0.043	0.000
C4	1.3B	Checkpoint (tokens, z)	0.017	0.002	0.013	0.020	1.000

Corpus	Size	Term	Mean	SD	2.5%	97.5%	$P(\beta > 0)$
C4	1.3B	Overall freq (ln total, z within ckpt)	0.232	0.011	0.210	0.252	1.000

Table 7: Posterior fixed effects for absolute signed gap. Mean = posterior mean; SD = posterior SD; 2.5%/97.5% = 95% credible interval bounds.